CHUNKING MEMORY VS RECALL

TITLE:

Design a User Interface where users recall visual social media icons and evaluate the effect of chunking on memory performance.

AIM:

To design and prototype a memory recall task using chunked social media icons and analyze the effect of chunking on short-term memory performance

SOFTWARE USED:

• Figma (UI Design Tool)

THEORY:

Memory recall refers to the ability of an individual to retrieve previously viewed information from short-term memory. Human memory has a limited capacity, meaning only a small number of items can be retained for a short duration.

Chunking is a memory improvement technique where individual elements are grouped into meaningful units. According to Miller's Law (7 ± 2 rule), a person can remember approximately 5 to 9 items in short-term memory.

In this experiment, five familiar social media icons (Facebook, Instagram, WhatsApp, Messenger, Skype) are visually grouped in a single container. Since the icons are commonly recognized and presented together, chunking helps users encode them efficiently and improves recall accuracy.

This experiment evaluates how grouped visual elements influence short-term memory recall.

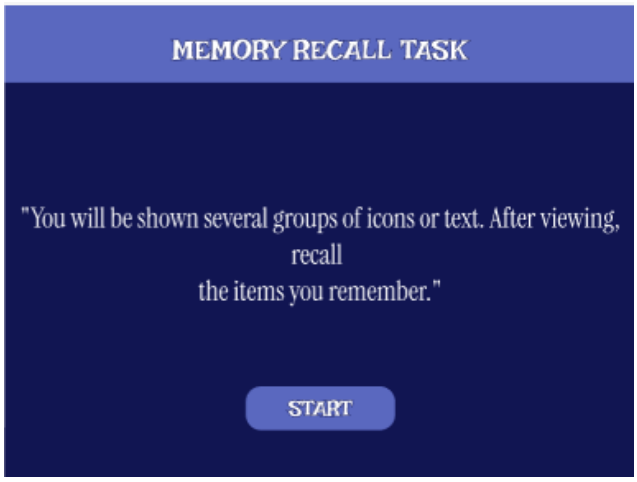
DESIGN DESCRIPTION:

A. Home Screen

The Home Screen contains:

• Title: “MEMORY RECALL TASK”
• Instruction text explaining the experiment
• Clear description of recalling viewed items
• “START” button

Purpose:
To provide clear instructions before beginning the memory experiment.



B. Viewing / Chunking Phase

The Viewing Phase displays:

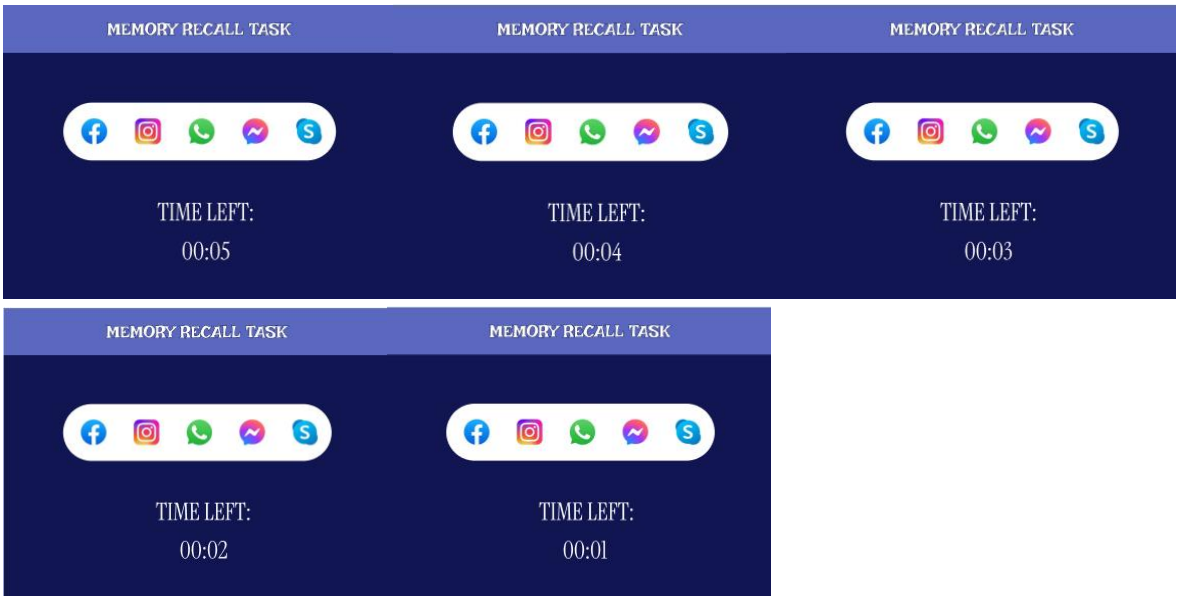
• Five social media icons:

- Facebook
- Instagram
- WhatsApp
- Messenger
- Skype

• Icons grouped inside a rounded white container
• Countdown timer (5 → 1 seconds)
• “TIME LEFT” label

Purpose:
To allow users to observe and encode grouped visual information into short-term memory within 5 seconds.

The countdown simulates a controlled memory exposure duration.

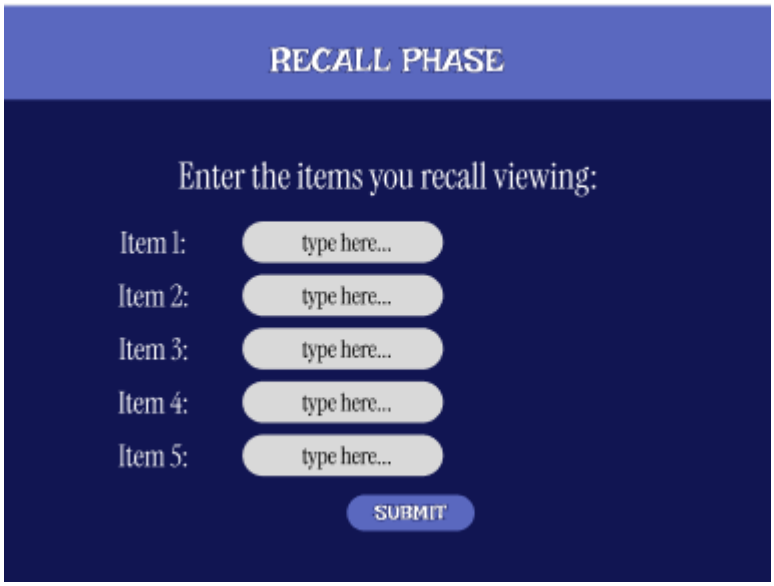


C. Recall Phase

The Recall Screen includes:

• Title: “RECALL PHASE”
• Instruction: “Enter the items you recall viewing”
• Five input fields labeled Item 1 to Item 5
• “SUBMIT” button

Purpose:
To measure how accurately users can recall the icons after the viewing phase.

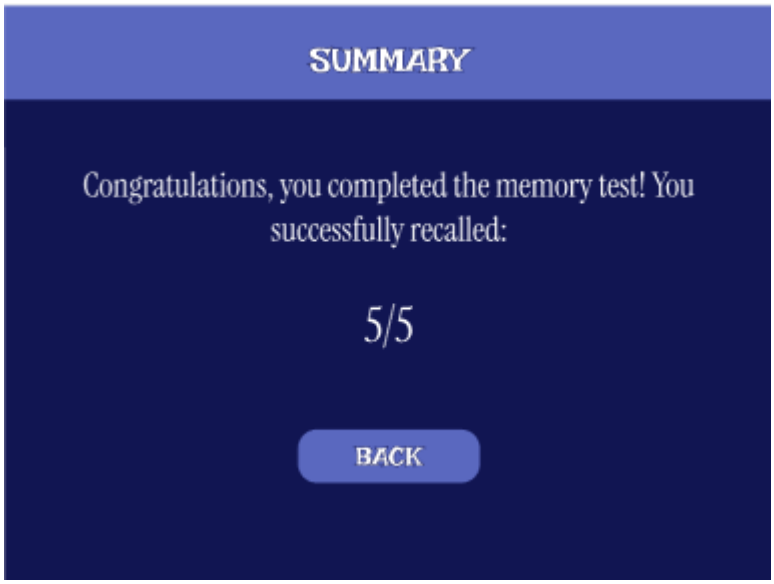


D. Summary / Result Screen

The Summary Screen displays:

• Title: “SUMMARY”
• Feedback message
• Final score (example: 5/5)
• “BACK” button

Purpose:
To provide performance feedback and show recall accuracy.



PROCEDURE:

1. Created a Home Screen with instructions and a Start button.
2. Designed the Viewing Phase displaying five grouped social media icons.
3. Added a countdown timer (5 seconds) using prototype delay interaction in Figma.
4. After the timer ends, navigated automatically to the Recall Phase.
5. Designed input fields for users to enter recalled items.

6. Added a Submit button linked to the Summary Screen.

7. Displayed the final recall score (e.g., 5/5 items correctly recalled).

OBSERVATION:

Users were able to recall most or all of the icons because:

• The icons were visually grouped in one container.
• The brands are familiar and recognizable.
• Chunking reduced cognitive load.
• The structured layout helped visual encoding.

Most users achieved high recall accuracy (e.g., 5/5).

RESULT:

The experiment demonstrates that chunking familiar visual elements (social media icons) significantly improves short-term memory recall performance.

CONCLUSION:

Chunking enhances memory retention by organizing information into structured and meaningful visual groups.

This experiment confirms that grouped visual elements improve recall accuracy compared to unorganized presentation.

The structured UI design, timer-based exposure, and recall phase effectively simulate short-term memory testing.